

A novel mathematical model of ATM/p53/NF- $\kappa$ B pathways points to the importance of the DDR switch-off mechanisms

## ADDITIONAL FILE

### Numerical implementation

Our numerical implementation of the hybrid algorithm following Haseltine-Rawlings postulate [1] can be presented in the following steps:

- 1 For the given time  $t$  and given values of all variables (deterministic and stochastic), and knowing both input values, calculate the total propensity function  $a^*(t)$  of the occurrence of any possible reaction:

$$\begin{aligned}
 a^*(t) = & a_1 + a_2 + a_3(M - Ra(t)) + a_4Ra(t) + \\
 & + a_5(NA_{atm} - G_{atm}(t)) + a_6G_{atm}(t) + \\
 & + a_7(NA_{chk2} - G_{chk2}(t)) + a_8G_{chk2}(t) + \\
 & + a_9(NA_{wip1} - G_{wip1}(t)) + a_{10}G_{wip1}(t) + \\
 & + a_{11}(NA_{p53} - G_{p53}(t)) + a_{12}G_{p53}(t) + \\
 & + a_{13}(NA_{ikba} - G_{ikba}(t)) + a_{14}G_{ikba}(t) + \\
 & + a_{15}(NA_{a20} - G_{a20}(t)) + a_{16}G_{a20}(t) + \\
 & + a_{17}(NA_{bax} - G_{bax}(t)) + a_{18}G_{bax}(t) + \\
 & + a_{19}(NA_{p21} - G_{p21}(t)) + a_{20}G_{p21}(t) + \\
 & + a_{21}(NA_{mdm2} - G_{mdm2}(t)) + \\
 & + a_{22}G_{mdm2}(t) + a_{23}(NA_{pten} - G_{pten}(t)) + \\
 & + a_{24}G_{pten}(t).
 \end{aligned} \tag{1}$$

where  $NA_i$  denotes total number of “i-th species”,  $M$  denotes total number of the cell receptors,  $Ra$  denotes the number of active receptors at the time  $t$  and  $G_i$  “i-th’ species” gene state at the time  $t$ .

- 2 Select two random numbers  $p_1$  and  $p_2$  from the uniform distribution on  $[0, 1]$ .
- 3 Using the fourth order Runge-Kutta solver evaluate the system of ODEs, accounting for fast reactions, until the time  $t + \tau$  such that

$$\ln(p_1) + \int_t^{t+\tau} a^*(s)ds = 0. \tag{2}$$

- 4 Determine which one of the possible slow reactions occurs at time  $t + \tau$  using the inequality

$$\sum_{i=1}^{k-1} a_i(t + \tau) < p_2 a^*(t + \tau) \leq \sum_{i=1}^k a_i(t + \tau). \tag{3}$$

where  $k$  is the index of the reaction to occur.

- 5 Replace the time  $t$  with  $t + \tau$  and if the simulation horizon  $T$  is not reached go back to step 1.

To find the time  $\tau$  from Eq. 1 we regularly sample our ODE system each  $\Delta t = 0.1$  sec to determine if the stochastic reaction occurs in the period of time from the last check-time to current one. Of course, in this method some inaccuracies may appear in the stochastic events “firings” the time. Although, because the gene switching as well as receptors activation/inactivation and DSB occurrence/repair times are in the order of minutes, i.e. much larger then the chosen sampling time,  $\Delta t$  inaccuracies should be rare and should not affect the final results.

#### References

1. Haseltine EL, Rawlings JB. Approximate simulation of coupled fast and slow reactions for stochastic chemical kinetics. J Chem Phys. 2002;117:6959–6969.